Project Design Phase
Proposed Solution Template

Date	05 July 20265
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	**Parameter**	**Description**
	Problem Statement (Problem to be solved)	Problem Statement (Problem to be solved) Electric vehicle (EV) adoption is hindered by "range anxiety" and uncertainty regarding the availability of charging infrastructure. Drivers currently struggle to plan long trips and optimize their routes based on real-time battery life and the status of charging networks, which leads to inefficiencies and increased driver anxiety.
	Idea / Solution description	The proposed solution is an advanced, interactive visualization tool. It integrates real-time vehicle telematics—such as State of Charge (SoC) and battery health—with live charging station data, traffic information, and topographical data. The tool provides predictive range modeling and smart route planning with optimized charging stops, all visualized on a user-friendly, map-based dashboard.
	Novelty / Uniqueness	The uniqueness of this solution lies in its predictive algorithms, which combine vehicle data with dynamic environmental factors (such as traffic and elevation) and live charging infrastructure status to offer comprehensive, real-time range and charge visualization. It also employs a multi-layered visualization approach, allowing for custom user views, such as "time to next charge" or "charge cost optimization".
	Social Impact / Customer Satisfaction	This tool increases consumer confidence in EVs, thereby reducing range anxiety and encouraging the adoption of sustainable transportation. It enhances the overall EV ownership experience by simplifying trip planning and optimizing travel time. Additionally, it promotes efficient use of the power grid by encouraging off-peak charging through its cost-visualization features.

	Business Model (Revenue Model)	The dashboard can be offered as a B2B SaaS product to automotive OEMs (as a value-added feature for their infotainment systems) and to fleet management companies. A B2C version could be offered as a premium subscription mobile app. Further potential revenue streams include data partnerships with charging network providers and navigation companies.
	Scalability of the Solution	The platform is highly scalable and capable of integrating data from diverse EV models and charging networks globally. It can be expanded to include predictive maintenance insights and integrate with smart grid systems for Vehicle-to-Grid (V2G) optimization. The underlying data model can also be adapted for other electric mobility solutions, such as e-bikes and e-buses.